

Set-A



Midterm Examination

Full Marks: **10 x 2 + 15 = 35**
Time: **1 hour 20 minutes**
Date: **22 November 2025**

Student ID:	Name:	Section:
--------------------	--------------	-----------------

[Answer all **THREE** questions. **Return** the question paper with the answer script.]

Question 1: [CO2]

You are tasked to design a circuit that determines the **parity** (even/odd) of **consecutive occurrences** of the sequence “**100**” in the input stream. The system has a single bit input w and two outputs z_1 and z_2 . z_1 should be high if an **odd** number of ‘100’s are detected whereas z_2 should be high if an **even** number of ‘100’ are detected (**consecutively**). The table below shows a sample operation:

clk	t0	t1	t2	t3	t4	t5	t6	t7	t8	t9	t10	t11	t12	t13	t14	t15	t16	t17
w	0	0	1	0	0	0	0	1	0	0	1	0	0	0	1	0	0	0
z1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1
z2	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Before t13, '100' appeared twice in a row, making $z_2=1$.

(a)	Design the state diagram for the corresponding FSM. How many Flip-Flips are required to implement in a circuit?	[5]
(b)	Derive the State Assigned Table using Binary Encoding .	[2]
(c)	Find the Boolean expression for the output logic using KMAP.	[3]

Question 2: [CO2]

Simon is developing a mechanism for his remote-controlled robot to automatically detect and respond to real fire incidents by activating a water-sprinkling system. The system integrates two sensors, a smoke detector and a heat sensor, to ensure reliable fire detection.

The *smoke detector* operates in an **active-low** configuration, where its output pin (**x**) becomes low (0) when the detected smoke concentration exceeds a defined threshold (100 ppm); otherwise, it remains high (1). Conversely, the *heat sensor* works in an **active-high** configuration, with its output pin (**y**) turning high (1) when the ambient temperature rises above a preset limit (50°C); otherwise, it remains low (0).

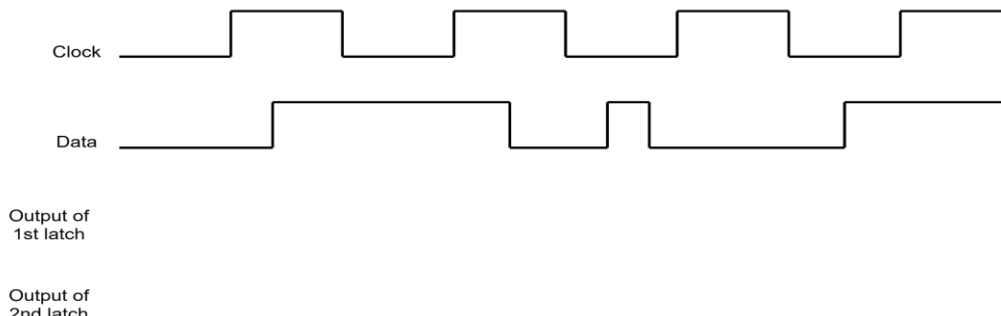
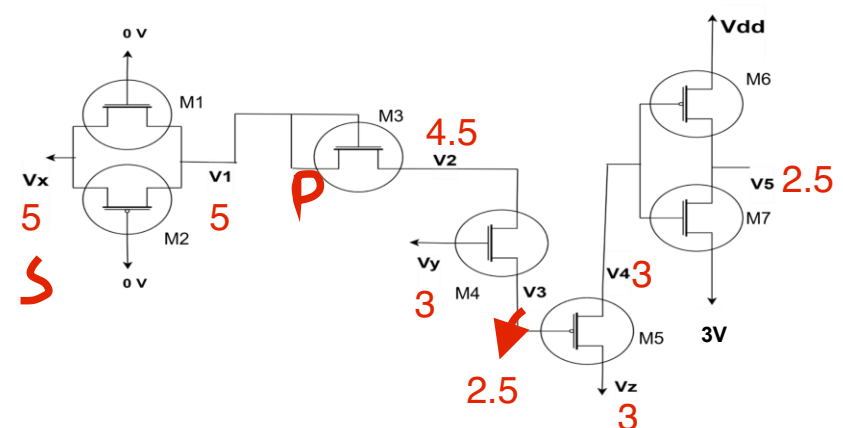
The water sprinkler system is activated only when the heat sensor detects a high temperature **for two consecutive** clock cycles, followed by the simultaneous detection of **both high temperature and high smoke concentration in the third cycle** at a particular location. This timing-based logic ensures accurate fire detection while minimizing false activations.

A summary of the operational logic is presented in the following table:

[illegible]

(a)	Determine the type of FSM that would be needed for the given problem and design the state diagram for the corresponding FSM.	[2]
(b)	Derive the state-assigned table using the Binary encoding from your state diagram in (a).	[2]
(c)	Find the expression for the next state and the output for the FSM using KMAP.	[3]

Question 3: [CO1]

(a)	i) Write down the truth table and output logic function “Y” of a 4 to 1 mux whose input lines are I0, I1, I2, I3 and select pins are S1, S0. ii) Implement the mux logic function with Transmission gate. iii) Compare the number of transistors needed with the conventional CMOS logic design approach. (Assume all the input's inverted form are available.)	[2+2+1]
(b)	Suppose you have been given negative level D latches, system clock signal and input data signal. i) Design a negative edge D flip-flop showing the clock and data signal flow through the given latch/latches. (You need not show the circuit diagram inside latch.) ii) Draw the output of the latches for the clock and data signal shown below for your design in b(i). (Draw on the question paper.) 	[2+3]
(c)	Find the node voltages for the circuit shown in the figure below analyzing pass transistor DC characteristics for $V_x = 5V$, $V_y = 3V$, $V_z = 3V$, $V_{dd} = 5V$. Assume $V_{tn} = 0.5V$, $V_{tp} = 0.3V$. [M1,M3,M4,M7 are NMOS and M2,M5,M6 are PMOS pass transistor] 	[5]